

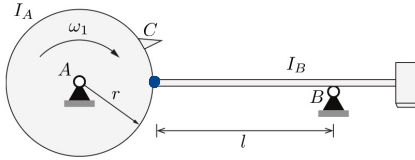
Problems Week 8

~lschen

Problem 1: Hammer Mechanism

A hammer mill consists of a uniform disk of radius r with moment of inertia I_A hinged at point A and a hammer with moment of inertia I_B that is hinged at a distance l from the disk. The disk rotates initially with an angular velocity ω_1 until its thumb C hits the shaft of the hammer. The coefficient of restitution of this impact is a known value e . The hammer is at rest before the impact. You may assume that the thumb C is of negligible mass and dimension. Neglect gravity.

Given: $I_A, I_B, r, l, e, \omega_1$



(a) Determine the angular velocity of the disk and hammer immediately after the impact.

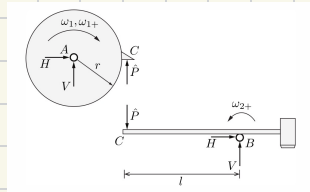
(b) Determine the dissipated (lost) energy for $I_B/I_A = l^2/r^2$.

a)

collision of rigid bodies: linear and angular momentum balance with contact point S:

$$\begin{aligned} \hat{P} &= m_1 [v_{CM1}(t_+) - v_{CM1}(t_-)], & r_{CM1S} \times \hat{P} &= I_{CM1} [\omega_1(t_+) - \omega_1(t_-)] \\ -\hat{P} &= m_2 [v_{CM2}(t_+) - v_{CM2}(t_-)], & r_{CM2S} \times (-\hat{P}) &= I_{CM2} [\omega_2(t_+) - \omega_2(t_-)] \end{aligned}$$

$$\begin{aligned} \text{disc: } \begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ \hat{P} \\ 0 \end{pmatrix} &= \underline{I}_A [\omega_1(t_+) - \omega_1(t_-)] \\ r \hat{P} e_3 &= I_A (\omega_1^+ - \omega_1) \quad (I) \end{aligned}$$



$$\begin{aligned} \text{hammer: } \begin{pmatrix} -l \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ \hat{P} \\ 0 \end{pmatrix} &= \underline{I}_B [\omega_2^+ - \omega_2^-] \\ l \hat{P} e_3 &= I_B \omega_2^+ e_3 \quad (II) \end{aligned}$$

from:

coefficient of restitution for a two-body collision (contact points S_1 and S_2):

$$e = -\frac{v_{S2}^u(t_+) - v_{S1}^u(t_+)}{v_{S2}^u(t_-) - v_{S1}^u(t_-)}$$

$$e = \frac{v_{c1}^+ - v_{c2}^+}{v_{c1}^- - v_{c2}^-} \quad \text{e is known, subscript c is at impact point}$$

$$e = \frac{v_{c1}^+ - v_{c2}^+}{\omega_1 r - 0} = \frac{\omega_1^+ r - \omega_2^+ l}{\omega_1 r - 0} \quad (III)$$

We have 3 unknowns $(\hat{P}, \omega_1^+, \omega_2^+)$ & 3 eq.

$$\begin{aligned} \text{from I \& II: } \omega_2^+ &= \frac{l I_A}{I_B} (\omega_1 - \omega_1^+) \\ \text{from III: } \omega_1^+ &= e \omega_1 + \omega_2^+ \frac{l}{r} \end{aligned} \quad \left. \begin{array}{l} \text{by doing some} \\ \text{ugly math:} \end{array} \right\} \quad \underline{\underline{\omega_1^+ = \omega_1 \frac{1 - \frac{r^2}{l^2} \frac{I_B}{I_A}}{1 + \frac{r^2}{l^2} \frac{I_B}{I_A}}}} \quad \underline{\underline{\omega_2^+ = \omega_1 \frac{(1+e)r}{1 + \frac{r^2}{l^2} \frac{I_B}{I_A}}}}$$

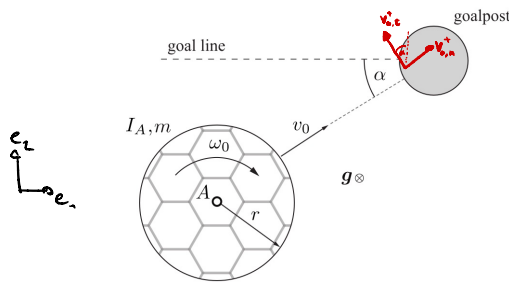
$$\text{b) find } \Delta T = \frac{1}{2} I_A \omega_1^2 - \left(\frac{1}{2} I_A \omega_1^{+2} + \frac{1}{2} I_B \omega_2^{+2} \right) \quad \text{using the given relation and plugging in for } \omega_1^+ \& \omega_2^+ \rightarrow \Delta T = \frac{1}{4} (1-e^2) I_A \omega_1^2$$

Good to know: if $e = 1$ (=no dissipated energy) we have $\omega_1^+ = 0$ & $\omega_2^+ = \omega_1 \frac{r}{l}$ (due to given relation)
= T conserved

Problem 2: The Decisive Shot

A soccer ball of mass m and moment of inertia I_A hits a *rough* goalpost *central* with initial velocity v_0 under an angle α to the goal line, with a known coefficient of restitution e . Assume that the goalpost is immovable and rigid. Gravity acts perpendicular to the plane, as indicated. Assume that the ball *sticks* to the rough goalpost during the collision.

Given: $I_A, m, r, \alpha, e, v_0, g$



What is the minimum required initial spin (i.e., the minimum initial angular velocity ω_0) of the ball to cross the goal line (assuming that it eventually crosses the goal line if it is deflected by some angle greater than α)?

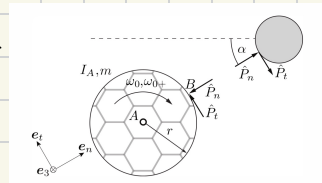
Condition is: $v(t_+) \cdot e_1 > 0$

a)

collision of rigid bodies: linear and angular momentum balance with contact point S:

$$\begin{aligned} \hat{P} &= m_1 [v_{CM1}(t_+) - v_{CM1}(t_-)], & r_{CM1S} \times \hat{P} &= I_{CM1} [\omega_1(t_+) - \omega_1(t_-)] \\ -\hat{P} &= m_2 [v_{CM2}(t_+) - v_{CM2}(t_-)], & r_{CM2S} \times (-\hat{P}) &= I_{CM2} [\omega_2(t_+) - \omega_2(t_-)] \end{aligned}$$

FBD:



Body A: $m(v_{0,t}^+ - v_0) = -\hat{P}_n e_n$

$$m v_{0,t}^+ = \hat{P}_t e_t$$

$$I_A (\omega_0^+ - \omega_0) e_3 = \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix} \times \begin{pmatrix} \hat{P}_t \\ -\hat{P}_n \\ 0 \end{pmatrix} = -r \hat{P}_t e_3$$

Body B: immovable

coefficient of restitution for a two-body collision (contact points S_1 and S_2):

$$e = -\frac{v_{S_2}^n(t_+) - v_{S_1}^n(t_+)}{v_{S_2}^n(t_-) - v_{S_1}^n(t_-)}$$

$$e = -\frac{v_{0,n}^+}{v_0} \Rightarrow -e v_0 = v_{0,n}^+$$

collision of two rough bodies, assuming no slip:

$$v_{S_1}^t(t_+) = v_{S_2}^t(t_+) \stackrel{!}{=} 0 \text{ since B immovable}$$

$$v_S^t = v_{0,t}^+ - r \omega_0^+ = 0 \quad (\text{rel. trans. form.})$$

Math: $I_A (\omega_0^+ - \omega_0) = -r \hat{P}_t$

$$\hat{P}_t = m v_{0,t}^+$$

$$\rightarrow \omega_0^+ = -\frac{r}{I_A} \hat{P}_t + \omega_0$$

$$= -\frac{r}{I_A} m v_{0,t}^+ + \omega_0$$

$$\rightarrow v_{S,t}^+ = 0 = v_{0,t}^+ - r \omega_0^+$$

$$0 = v_{0,t}^+ + \frac{r^2}{I_A} m v_{0,t}^+ - r \omega_0$$

$$v_{0,t}^+ = \frac{r \omega_0}{1 + \frac{r^2}{I_A} m}$$

Geometric condition for ball to cross the line : $v_{0,n}^+ \sin \alpha + v_{0,t}^+ \cos \alpha > 0$

$$-e v_0 \sin \alpha + \frac{r \omega_0}{1 + \frac{r^2 m}{I_A}} \cos \alpha > 0$$

$$\omega_0 > \frac{e v_0}{r} \tan \alpha \left(1 + \frac{r^2 m}{I_A} \right)$$